

PRESIDENCY UNIVERSITY

School of Information Science
Department of Master of Computer Applications

AI-Powered Personalized Learning Path
Nelotsavam Platform — Complete Source Code

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FRONTEND

Next.js 15	React 19	Tailwind CSS	NextAuth	Zustand	TypeScript
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BACKEND

Node.js	Express.js	MongoDB	Mongoose	JWT	Socket.io
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Table of Contents

■ PART A FRONTEND (Next.js 15 + React 19 + TypeScript)

- A1 Project Configuration
- A2 Root Layout & Global Styles
- A3 Home Page (Landing)
- A4 Authentication Pages (Login/Register)
- A5 Dashboard Page
- A6 Marketplace Page
- A7 Community Forum Page
- A8 Learning Center Page
- A9 NextAuth API Route
- A10 Zustand Global Store
- A11 API Client Library
- A12 Providers Component

■ PART B BACKEND (Node.js + Express.js + MongoDB)

- B1 Project Configuration
- B2 Application Entry Point (index.js)
- B3 Auth Middleware
- B4 Auth Module (Register / Login)
- B5 Friends Module
- B6 Water Intake Module
- B7 Activity Session Module
- B8 Order Module
- B9 Unit Tests
- C1 Full API Endpoint Reference
- C2 Project Setup Instructions

■ PART A — FRONTEND (Next.js 15 · React 19 · TypeScript)

A1. Project Configuration

■ package.json

```
{ "name": "nelotsavam-frontend", "version": "1.0.0", "private": true, "scripts": { "dev": "next dev", "build": "next build", "start": "next start", "lint": "next lint" }, "dependencies": { "next": "15.0.0", "react": "19.0.0", "react-dom": "19.0.0", "next-auth": "^4.24.5", "zustand": "^4.4.7", "@radix-ui/react-dialog": "^1.0.5", "@radix-ui/react-dropdown-menu": "^2.0.6", "@radix-ui/react-tabs": "^1.0.4", "@radix-ui/react-toast": "^1.1.5", "@radix-ui/react-avatar": "^1.0.4", "@radix-ui/react-progress": "^1.0.3", "class-variance-authority": "^0.7.0", "clsx": "^2.0.0", "tailwind-merge": "^2.1.0", "lucide-react": "^0.294.0", "axios": "^1.6.2", "date-fns": "^2.30.0", "@prisma/client": "^5.6.0" }, "devDependencies": { "typescript": "^5.3.2", "@types/node": "^20.10.0", "@types/react": "^18.2.39", "@types/react-dom": "^18.2.17", "tailwindcss": "^3.3.6", "autoprefixer": "^10.4.16", "postcss": "^8.4.32", "prisma": "^5.6.0", "eslint": "^8.54.0", "eslint-config-next": "15.0.0" } }
```

■ next.config.js

```
/** @type {import('next').NextConfig} */ const nextConfig = { experimental: { appDir: true }, images: { domains: ["res.cloudinary.com", "avatars.githubusercontent.com"], }, }; module.exports = nextConfig;
```

■ tailwind.config.ts

```
import type { Config } from "tailwindcss"; const config: Config = { content: [ "./pages/**/*.{js,ts,jsx,tsx,mdx}", "./components/**/*.{js,ts,jsx,tsx,mdx}", "./app/**/*.{js,ts,jsx,tsx,mdx}", ], theme: { extend: { colors: { primary: { DEFAULT: "#1a237e", light: "#534bae", dark: "#000051" }, secondary: { DEFAULT: "#2e7d32", light: "#60ad5e", dark: "#005005" }, accent: { DEFAULT: "#f9a825", light: "#ffd95a", dark: "#c17900" }, }, fontFamily: { sans: ["Inter", "sans-serif"] }, }, }, plugins: [], }; export default config;
```

A2. Root Layout

■ app/layout.tsx

```
import type { Metadata } from "next"; import { Inter } from "next/font/google"; import "../globals.css";
import { Providers } from "@components/layout/Providers"; import { Toaster } from "@components/ui/toaster";
const inter = Inter({ subsets: ["latin"] }); export const metadata: Metadata = { title: "Nelotsavam -
Empowering Kerala's Rice Farmers", description: "A gamified digital platform for Kerala rice farmers
featuring community tools, " + "expert consultation, marketplace, and sustainable farming rewards.", };
export default function RootLayout({ children, }: { children: React.ReactNode; }) { return ( <html lang="en">
<body className={inter.className}> <Providers> {children} <Toaster /> </Providers> </body> </html> ); }
```

A3. Home Page (Landing)

■ app/page.tsx

```
"use client"; import Link from "next/link"; import { ArrowRight, Award, Users, ShoppingBag, BookOpen, CloudSun, Star } from "lucide-react"; const features = [ { icon: Award, title: "Gamified Learning", desc: "Earn Dhanya points & badges through Sevas and farming milestones." }, { icon: Star, title: "Sustainable Practices", desc: "Learn and adopt eco-friendly cultivation techniques for Kerala." }, { icon: Users, title: "Farmer Community", desc: "Connect with fellow farmers, share knowledge, and grow together." }, { icon: BookOpen, title: "Expert Guidance", desc: "Consult agricultural experts on crops, irrigation, and pest control." }, { icon: CloudSun, title: "Local Weather & Insights", desc: "Get real-time weather updates tailored to your farming location." }, { icon: ShoppingBag, title: "Marketplace", desc: "Buy and sell agri-products directly with verified agri-businesses." }, ]; export default function HomePage() { return ( <main className="min-h-screen bg-gradient-to-br from-green-50 to-indigo-50"> { /* Nav */ <nav className="bg-white shadow-sm sticky top-0 z-50"> <div className="max-w-7xl mx-auto px-4 py-3 flex items-center justify-between"> <div className="flex items-center gap-2"> <span className="text-2xl font-bold text-primary-DEFAULT">■ Nelotsavam</span> </div> <div className="flex items-center gap-4"> <button className="text-sm text-gray-600 hover:text-primary-DEFAULT">English</button> <button className="text-sm text-gray-600 hover:text-primary-DEFAULT">Malayalam</button> <Link href="/auth/login" className="bg-primary-DEFAULT text-white px-4 py-2 rounded-lg text-sm font-medium hover:bg-primary-dark transition"> Join Now </Link> </div> </div> { /* Hero */ <section className="max-w-7xl mx-auto px-4 py-20 text-center"> <p className="text-sm font-medium text-secondary-DEFAULT bg-green-100 inline-block px-3 py-1 rounded-full mb-4"> ■ Digital Harvest Festival </p> <h1 className="text-5xl font-bold text-gray-900 mb-6 leading-tight"> Empowering Kerala's <br /> <span className="text-primary-DEFAULT">Rice Farmers</span> </h1> <p className="text-xl text-gray-600 max-w-2xl mx-auto mb-10"> Join thousands of farmers in a gamified journey of sustainable rice cultivation. Learn traditional methods, earn rewards, and build community connections. </p> <div className="flex items-center justify-center gap-4"> <Link href="/auth/register" className="bg-primary-DEFAULT text-white px-8 py-3 rounded-xl font-semibold flex items-center gap-2 hover:bg-primary-dark transition shadow-lg"> Start Your Journey <ArrowRight size={18} /> </Link> <Link href="/dashboard" className="border border-primary-DEFAULT text-primary-DEFAULT px-8 py-3 rounded-xl font-semibold hover:bg-indigo-50 transition"> Learn More </Link> </div> </section> { /* Why Nelotsavam */ <section className="max-w-7xl mx-auto px-4 py-16"> <h2 className="text-3xl font-bold text-center text-gray-900 mb-12">Why Nelotsavam?</h2> <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-8"> {features.map(({ icon, title, desc }) => ( <div key={title} className="bg-white rounded-2xl p-6 shadow-sm hover:shadow-md transition border border-gray-100"> <div className="w-12 h-12 bg-indigo-100 rounded-xl flex items-center justify-center mb-4"> <Icon className="text-primary-DEFAULT" size={24} /> </div> <h3 className="text-lg font-semibold text-gray-900 mb-2">{title}</h3> <p className="text-gray-500 text-sm leading-relaxed">{desc}</p> </div> ))} </div> </section> { /* Footer */ <footer className="bg-primary-DEFAULT text-white mt-20 py-10"> <div className="max-w-7xl mx-auto px-4 text-center"> <p className="text-lg font-bold mb-2">■ Nelotsavam</p> <p className="text-indigo-200 text-sm">Empowering Kerala's rice farming community through technology.</p> <p className="text-indigo-300 text-xs mt-4">© 2026 Nelotsavam · MCA Major Project · Presidency University, Bengaluru</p> </div> </footer> </main> ); }
```

A4a. Authentication – Login Page

■ app/auth/login/page.tsx

```
"use client"; import { useState } from "react"; import { signIn } from "next-auth/react"; import Link from "next/link"; import { Eye, EyeOff, Loader2 } from "lucide-react"; export default function LoginPage() { const [email, setEmail] = useState(""); const [password, setPassword] = useState(""); const [show, setShow] = useState(false); const [loading, setLoading] = useState(false); const [error, setError] = useState(""); const handleSubmit = async (e: React.FormEvent) => { e.preventDefault(); setLoading(true); setError(""); const res = await signIn("credentials", { email, password, redirect: false }); if (res?.error) setError("Invalid email or password."); else window.location.href = "/dashboard"; setLoading(false); }; return ( <div className="min-h-screen bg-gradient-to-br from-green-50 to-indigo-50 flex items-center justify-center px-4"> <div className="bg-white rounded-3xl shadow-xl w-full max-w-md p-8"> <div className="text-center mb-8"> <p className="text-3xl mb-2">■</p> <p className="text-xs font-medium text-green-600 bg-green-50 inline-block px-3 py-1 rounded-full mb-2"> Digital Harvest Festival </p> <h1 className="text-2xl font-bold text-gray-900">Welcome Back</h1> <p className="text-sm text-gray-500 mt-1">Sign in to continue your farming journey</p> </div> <form onSubmit={handleSubmit} className="space-y-4"> <div> <label className="text-sm font-medium text-gray-700 block mb-1">Email Address</label> <input type="email" value={email} onChange={e => setEmail(e.target.value)} required placeholder="farmer@example.com" className="w-full px-4 py-3 border border-gray-200 rounded-xl text-sm focus:outline-none focus:ring-2 focus:ring-primary-DEFAULT" /> </div> <div> <label className="text-sm font-medium text-gray-700 block mb-1">Password</label> <div className="relative"> <input type={show ? "text" : "password"} value={password} onChange={e => setPassword(e.target.value)} required placeholder="Create a strong password" className="w-full px-4 py-3 border border-gray-200 rounded-xl text-sm focus:outline-none focus:ring-2 focus:ring-primary-DEFAULT pr-10" /> <button type="button" onClick={() => setShow(!show)} className="absolute right-3 top-1/2 -translate-y-1/2 text-gray-400"> {show ? <EyeOff size={18} /> : <Eye size={18} />} </button> </div> </div> {error && <p className="text-sm text-red-600 bg-red-50 px-4 py-2 rounded-xl">{error}</p>} <button type="submit" disabled={loading} className="w-full bg-primary-DEFAULT text-white py-3 rounded-xl font-semibold flex items-center justify-center gap-2 hover:bg-primary-dark transition disabled:opacity-60"> {loading && <Loader2 size={16} className="animate-spin" />} Sign In </button> </form> <p className="text-center text-sm text-gray-500 mt-6"> Don't have an account?{" "} <Link href="/auth/register" className="text-primary-DEFAULT font-semibold hover:underline">Sign Up</Link> </p> <p className="text-center mt-2"> <Link href="/" className="text-xs text-gray-400 hover:underline">← Back to Home</Link> </p> </div> </div> ); }
```

A4b. Authentication – Register Page

■ app/auth/register/page.tsx

```
"use client"; import { useState } from "react"; import Link from "next/link"; import { Loader2 } from
"lucide-react"; import axios from "axios"; const districts = ["Alappuzha","Ernakulam","Idukki","Kannur","Kas
aragod","Kollam","Kottayam","Kozhikode","Malappuram","Palakkad","Pathanamthitta","Thiruvananthapuram","Thris
sur","Wayanad"]; const farmSizes = ["Less than 0.5 acres","0.5-1 acre","1-2 acres","2-5 acres","5+ acres"];
const experienceLevels = ["Beginner (0-2 years)","Intermediate (3-7 years)","Experienced (8-15
years)","Expert (15+ years)"]; export default function RegisterPage() { const [form, setForm] = useState({
fullName:"", phone:"", email:"", district:"", farmSize:"", experience:"", password:"", confirm:"" }); const
[loading, setLoading] = useState(false); const [error, setError] = useState(""); const set = (k: string) =>
(e: React.ChangeEvent<HTMLInputElement | HTMLSelectElement>) => setForm(f => ({ ...f, [k]: e.target.value
})); const handleSubmit = async (e: React.FormEvent) => { e.preventDefault(); if (form.password !==
form.confirm) { setError("Passwords do not match."); return; } setLoading(true); setError(""); try { await
axios.post("/api/auth/register", form); window.location.href = "/auth/login"; } catch (err: any) {
setError(err.response?.data?.message ?? "Registration failed."); } finally { setLoading(false); } } const
field = (label: string, key: string, type = "text", placeholder = "") => ( <div> <label className="text-sm
font-medium text-gray-700 block mb-1">{label} </label> <input type={type} value={form[key]}
onChange={set(key)} required placeholder={placeholder} className="w-full px-4 py-2.5 border border-gray-200
rounded-xl text-sm focus:outline-none focus:ring-2 focus:ring-primary-DEFAULT" /> </div> ); const select =
(label: string, key: string, opts: string[], placeholder: string) => ( <div> <label className="text-sm
font-medium text-gray-700 block mb-1">{label} </label> <select value={form[key]}
onChange={set(key)} required className="w-full px-4 py-2.5 border border-gray-200 rounded-xl text-sm
focus:outline-none focus:ring-2 focus:ring-primary-DEFAULT bg-white"> <option
value="">{placeholder}</option> {opts.map(o => <option key={o} value={o}>{o}</option>)} </select> </div> );
return ( <div className="min-h-screen bg-gradient-to-br from-green-50 to-indigo-50 flex items-center
justify-center px-4 py-10"> <div className="bg-white rounded-3xl shadow-xl w-full max-w-lg p-8"> <div
className="text-center mb-6"> <p className="text-3xl mb-2">■</p> <h1 className="text-2xl font-bold
text-gray-900">Create Your Account</h1>

<p className="text-sm text-gray-500 mt-1">Join the Digital Harvest Festival community</p> </div> <form
onSubmit={handleSubmit} className="space-y-4"> <div className="grid grid-cols-1 md:grid-cols-2 gap-4">
{field("Full Name", "fullName", "text", "Your full name")} {field("Phone Number", "phone", "tel", "+91
XXXXXXXXXX")} </div> {field("Email Address", "email", "email", "farmer@example.com")} <div className="grid
grid-cols-1 md:grid-cols-2 gap-4"> {select("District", "district", districts, "Select your district")}
{select("Farm Size", "farmSize", farmSizes, "Select farm size")} </div> {select("Farming Experience",
"experience", experienceLevels, "Select your experience level")} <div className="grid grid-cols-1
md:grid-cols-2 gap-4"> {field("Password", "password", "password", "Create a strong password")}
{field("Confirm Password", "confirm", "password", "Confirm your password")} </div> {error && <p
className="text-sm text-red-600 bg-red-50 px-4 py-2 rounded-xl">{error}</p> } <button type="submit"
disabled={loading} className="w-full bg-primary-DEFAULT text-white py-3 rounded-xl font-semibold flex
items-center justify-center gap-2 hover:bg-primary-dark transition disabled:opacity-60"> {loading && <Loader2
size={16} className="animate-spin" /> } Create Account </button> </form> <p className="text-center text-sm
text-gray-500 mt-6">Already have an account?{" "} <Link href="/auth/login" className="text-primary-DEFAULT
font-semibold hover:underline">Sign In</Link> </p> <p className="text-center mt-2"> <Link href="/"
className="text-xs text-gray-400 hover:underline">← Back to Home</Link> </p> </div> </div> ); }
```

A5. Dashboard Page

■ app/dashboard/page.tsx

```
"use client"; import { useSession } from "next-auth/react"; import { CheckCircle, Circle, Droplets, Users,
ShoppingBag, BookOpen, Award, TrendingUp } from "lucide-react"; const sevas = [ { title: "Check Water
Levels", points: 10, done: true }, { title: "Monitor Organic Pest Control", points: 15, done: true }, {
title: "Community Discussion", points: 20, done: false }, { title: "Learn Water Management", points: 25,
done: false }, ]; const quickActions = [ { icon: Users, label: "Add Expert", color: "bg-blue-100
text-blue-700" }, { icon: ShoppingBag, label: "Community Chat", color: "bg-green-100 text-green-700" }, {
icon: BookOpen, label: "Local Marketplace", color: "bg-orange-100 text-orange-700" }, { icon: Droplets,
label: "Learning Center", color: "bg-purple-100 text-purple-700" }, ]; export default function
DashboardPage() { const { data: session } = useSession(); const name = session?.user?.name ?? "Farmer";
return ( <div className="min-h-screen bg-gray-50"> { /* Top bar */ } <header className="bg-primary-DEFAULT
text-white px-6 py-4 flex items-center justify-between"> <div> <p className="text-sm text-indigo-200">Welcome
back,</p> <h1 className="text-xl font-bold">{name} ■</h1> <p className="text-xs text-indigo-200 mt-0.5">Your
farm is thriving! Keep up the great work.</p> </div> <div className="text-right text-sm"> <p
className="font-semibold">87/9 Tasks</p> </div> </header> <main className="max-w-7xl mx-auto px-4 py-6 grid
grid-cols-1 lg:grid-cols-3 gap-6"> { /* Left column */ } <div className="lg:col-span-2 space-y-6"> { /* Stats
*/ } <div className="grid grid-cols-2 md:grid-cols-4 gap-4"> [ { label: "Dhanya Points", value: "1,247",
icon: "■" }, { label: "Sevas Completed", value: "23", icon: "■" }, { label: "Community Helps", value: "#15",
icon: "■" }, { label: "Seva Level", value: "Level 5", icon: "■" }, ].map(({ label, value, icon }) => ( <div
key={label} className="bg-white rounded-2xl p-4 shadow-sm text-center"> <p className="text-2xl
mb-1">{icon}</p> <p className="text-xl font-bold text-gray-900">{value}</p> <p className="text-xs
text-gray-500 mt-1">{label}</p> </div> )) </div> )) </div>

{ /* Today's Sevas */ } <div className="bg-white rounded-2xl shadow-sm p-6"> <div className="flex items-center
justify-between mb-4"> <h2 className="font-bold text-gray-900">■ Today's Sevas (Tasks)</h2> <button
className="text-xs text-primary-DEFAULT font-medium">SEE ALL</button> </div> <div className="space-y-3">
{sevas.map((s) => ( <div key={s.title} className="flex items-center justify-between p-3 rounded-xl
bg-gray-50"> <div className="flex items-center gap-3"> {s.done ? <CheckCircle className="text-green-500"
size={20} /> : <Circle className="text-gray-300" size={20} /> } <div> <p className="text-sm font-medium
${s.done ? "line-through text-gray-400" : "text-gray-800"}"> {s.title} </p> <p className="text-xs
text-gray-400">+{s.points} pts</p> </div> </div> <span className={ `text-xs px-2 py-1 rounded-full font-medium
${s.done ? "bg-green-100 text-green-700" : "bg-gray-200 text-gray-600"} ` }> {s.done ? "Complete" : "Pending"}
</span> </div> ))) </div> </div> { /* Progress bars */ } <div className="bg-white rounded-2xl shadow-sm p-6">
<h2 className="font-bold text-gray-900 mb-4">■ Your Progress</h2> [ { label: "To Next Level (8)", value: 72
}, { label: "Weekly Seva Completion", value: 46 }, { label: "Community Participation", value: "High" as
unknown as number }, ].map(({ label, value }) => ( <div key={label} className="mb-4"> <div className="flex
justify-between text-sm mb-1"> <span className="text-gray-600">{label}</span> <span className="font-medium
text-gray-900">{typeof value === "number" ? `${value}%` : value}</span> </div> {typeof value === "number" &&
( <div className="h-2 bg-gray-100 rounded-full"> <div className="h-2 bg-primary-DEFAULT rounded-full"
style={{ width: `${value}%` }} /> </div> ) } </div> )) </div> </div> { /* Right column */ } <div
className="space-y-6"> { /* Weather */ } <div className="bg-gradient-to-br from-blue-500 to-blue-700 text-white
rounded-2xl p-5 shadow-sm">

<p className="text-sm font-medium opacity-80 mb-1">■■ Weather Today</p> <p className="text-4xl
font-bold">28°C</p> <p className="text-sm opacity-80 mt-1">Partly cloudy, good for harvest</p> <div
className="flex gap-4 mt-3 text-xs"> <span>■ Rain: 10%</span> <span>■ Humidity: 70%</span> </div> </div> { /*
Quick Actions */ }
```

A6. Marketplace Page

■ app/marketplace/page.tsx

```
"use client"; import { useState } from "react"; import { ShoppingCart, Star, Filter, Search } from
"lucide-react"; const categories = ["All Products", "Seeds", "Fertilizers", "Pest Control", "Harvest",
"Tools"]; const products = [ { id: 1, name: "Organic Neem Cake Fertilizer", seller: "Green Farm Supplies",
price: 450, unit: "kg", rating: 4.8, reviews: 234, badge: "Organic", orders: 1 }, { id: 2, name: "Premium
Jyothi Rice Seeds", seller: "Kerala Seed Co.", price: 320, unit: "kg", rating: 4.6, reviews: 189, badge:
"Popular", orders: 1 }, { id: 3, name: "Fresh Harvest Matta Rice", seller: "Farmer Direct", price: 1200,
unit: "quintal", rating: 4.9, reviews: 156, badge: "Organic", orders: 1 }, { id: 4, name: "Organic Pest
Control Spray", seller: "BioProtect Kerala", price: 280, unit: "litre", rating: 4.5, reviews: 97, badge:
"Bio", orders: 2 }, ]; export default function MarketplacePage() { const [active, setActive] = useState("All
Products"); const [search, setSearch] = useState(""); const filtered = products.filter(p =>
p.name.toLowerCase().includes(search.toLowerCase()) ); return ( <div className="min-h-screen bg-gray-50">
<header className="bg-primary-DEFAULT text-white px-6 py-4"> <h1 className="text-xl font-bold">■ Local
Marketplace</h1> <p className="text-sm text-indigo-200 mt-0.5">Buy & sell farming inputs and produce</p>
</header> <div className="max-w-7xl mx-auto px-4 py-6"> { /* Search & Filter */ } <div className="flex gap-3
mb-6"> <div className="flex-1 relative"> <Search className="absolute left-3 top-1/2 -translate-y-1/2
text-gray-400" size={18} /> <input value={search} onChange={e => setSearch(e.target.value)}
placeholder="Search products..." className="w-full pl-10 pr-4 py-2.5 border border-gray-200 rounded-xl
text-sm focus:outline-none focus:ring-2 focus:ring-primary-DEFAULT" /> </div> <button className="flex
items-center gap-2 px-4 py-2.5 border border-gray-200 rounded-xl text-sm bg-white"> <Filter size={16} />
Filter </button> </div> { /* Promo Banner */ } <div className="bg-gradient-to-r from-green-600 to-green-700
text-white rounded-2xl p-5 mb-6"> <p className="text-lg font-bold">■ Free Delivery Week!</p> <p
className="text-sm text-green-100">Get free delivery on all orders above ■500. Limited time offer!</p>
</div> { /* Categories */ } <div className="flex gap-2 overflow-x-auto pb-2 mb-6"> {categories.map(cat => (
<button
key={cat} onClick={() => setActive(cat)} className={`whitespace-nowrap px-4 py-2 rounded-full text-sm
font-medium transition ${ active === cat ? "bg-primary-DEFAULT text-white" : "bg-white text-gray-600 border
border-gray-200" }`} > {cat} </button> ))} </div> { /* Products Grid */ } <div className="grid grid-cols-1
sm:grid-cols-2 lg:grid-cols-3 xl:grid-cols-4 gap-5"> {filtered.map(p => ( <div key={p.id} className="bg-white
rounded-2xl shadow-sm overflow-hidden hover:shadow-md transition"> <div className="h-40 bg-gradient-to-br
from-green-100 to-green-200 flex items-center justify-center text-5xl"> ■ </div> <div className="p-4"> <div
className="flex items-start justify-between mb-1"> <span className="text-xs font-medium bg-green-100
text-green-700 px-2 py-0.5 rounded-full"> {p.badge} </span> <div className="flex items-center gap-1 text-xs
text-gray-500"> <Star size={12} className="text-yellow-400 fill-yellow-400" /> {p.rating} ({p.reviews})
</div> </div> <h3 className="text-sm font-semibold text-gray-900 mt-2">{p.name}</h3> <p className="text-xs
text-gray-500 mt-0.5">by {p.seller}</p> <div className="flex items-center justify-between mt-3"> <div> <p
className="text-lg font-bold text-gray-900">■{p.price}</p> <p className="text-xs text-gray-400">per
{p.unit}</p> </div> <button className="flex items-center gap-1.5 bg-primary-DEFAULT text-white px-3 py-2
rounded-xl text-xs font-medium hover:bg-primary-dark transition"> <ShoppingCart size={14} /> Buy Now
</button> </div> </div> </div> ))} </div> { /* Trust badges */ } <div className="grid grid-cols-3 gap-4 mt-8">
{[ { icon: "■", label: "Quality Assured", desc: "All products verified & certified" }, { icon: "■", label:
"Fast Delivery", desc: "Delivered within 2-3 days" }, { icon: "■", label: "Farmer Ratings", desc: "Reviewed
by real farmers" }, ].map(b => ( <div key={b.label} className="bg-white rounded-xl p-4 text-center
shadow-sm"> <p className="text-2xl mb-1">{b.icon}</p>
<p className="text-xs font-semibold text-gray-900">{b.label}</p> <p className="text-xs text-gray-400
mt-0.5">{b.desc}</p> </div> ))} </div> </div> </div> ); }
```

A7. Community Forum Page

■ app/community/page.tsx

```
"use client"; import { useState } from "react"; import { ThumbsUp, MessageSquare, Plus } from "lucide-react";
const trending = ["Organic Farming", "Pest Management", "Water Conservation", "Harvest Tips", "Soil Health"];
const posts = [ { id: 1, author: "Rajan Kumar", time: "2 hours ago", tag: "Organic Farming", title: "Best
organic fertilisers for rice fields", body: "I have been using neem cake and cow dung compost for my rice
fields. The results are amazing! The plants are healthier and pest problems have reduced significantly.",
likes: 45, comments: 12, }, { id: 2, author: "Saleeba Bai", time: "5 hours ago", tag: "Pest Management",
title: "Managing pests during monsoon", body: "Heavy rains are causing fungal infections in my rice crop. Has
anyone tried organic bio-pesticides? What worked for you in similar conditions?", likes: 28, comments: 8, },
{ id: 3, author: "Mohammed Ali", time: "1 day ago", tag: "Water Conservation", title: "Water-saving
techniques that work", body: "Implemented drip-and-spray (AWD) method in my fields. Successfully maintaining
optimal water levels, saving 30% more water while increasing yield.", likes: 61, comments: 15, }, ]; const
stats = [ { label: "Total Members", value: "12,456" }, { label: "Total Topics", value: "4,892" }, { label:
"Active Experts", value: "1,078" }, ]; export default function CommunityPage() { const [liked, setLiked] =
useState<number[]>([]); return ( <div className="min-h-screen bg-gray-50"> <header
className="bg-primary-DEFAULT text-white px-6 py-4 flex items-center justify-between"> <div> <h1
className="text-xl font-bold">■ Community Forum</h1> <p className="text-sm text-indigo-200">Share knowledge,
grow together</p> </div> <button className="flex items-center gap-2 bg-white text-primary-DEFAULT px-4 py-2
rounded-xl text-sm font-semibold"> <Plus size={16}> /> New Post </button> </header> <div className="max-w-7xl
mx-auto px-4 py-6 grid grid-cols-1 lg:grid-cols-3 gap-6"> /* Posts */ <div className="lg:col-span-2
space-y-4"> {posts.map(p => ( <div key={p.id} className="bg-white rounded-2xl shadow-sm p-5 hover:shadow-md
transition"> <div className="flex items-center gap-3 mb-3"> <div className="w-10 h-10 bg-indigo-100
rounded-full flex items-center justify-center font-bold text-primary-DEFAULT">

{p.author[0]} </div> <div> <p className="text-sm font-semibold text-gray-900">{p.author}</p> <p
className="text-xs text-gray-400">{p.time}</p> </div> <span className="ml-auto text-xs bg-green-100
text-green-700 px-2 py-0.5 rounded-full font-medium"> {p.tag} </span> </div> <h3 className="text-base
font-bold text-gray-900 mb-2">{p.title}</h3> <p className="text-sm text-gray-600 leading-relaxed
mb-4">{p.body}</p> <div className="flex items-center gap-4 border-t pt-3"> <button onClick={() => setLiked(l
=> l.includes(p.id) ? l.filter(x => x !== p.id) : [...l, p.id])} className={`flex items-center gap-1.5
text-sm transition ${liked.includes(p.id) ? "text-primary-DEFAULT font-semibold" : "text-gray-500"}`>
<ThumbsUp size={16}> /> {p.likes + (liked.includes(p.id) ? 1 : 0)} Likes </button> <button className="flex
items-center gap-1.5 text-sm text-gray-500"> <MessageSquare size={16}> /> {p.comments} Replies </button>
</div> </div> ))) </div> /* Sidebar */ <div className="space-y-5"> /* Stats */ <div className="bg-white
rounded-2xl shadow-sm p-5"> <h2 className="font-bold text-gray-900 mb-4">Community Stats</h2> {stats.map(s =>
( <div key={s.label} className="flex justify-between py-2 border-b last:border-0"> <span className="text-sm
text-gray-500">{s.label}</span> <span className="text-sm font-bold text-gray-900">{s.value}</span> </div>
))) </div> /* Trending */ <div className="bg-white rounded-2xl shadow-sm p-5"> <h2 className="font-bold
text-gray-900 mb-4">■ Trending Topics</h2> <div className="flex flex-wrap gap-2"> {trending.map(t => ( <span
key={t} className="text-xs bg-indigo-50 text-primary-DEFAULT px-3 py-1.5 rounded-full font-medium
cursor-pointer hover:bg-indigo-100 transition"> {t} </span> ))) </div> </div> /* Guidelines */ <div
className="bg-white rounded-2xl shadow-sm p-5"> <h2 className="font-bold text-gray-900 mb-3">Community
Guidelines</h2>

{["Be respectful to all farmers", "Share accurate information", "Use appropriate language", "Help fellow
farmers grow", "Report inappropriate content"].map(g => ( <p key={g} className="text-xs text-gray-500 mb-2">•
{g}</p> ))) </div> </div> </div> </div> ); }
```

A8. Learning Center Page

■ app/learning/page.tsx

```
"use client"; import { BookOpen, Clock, Users, Award, Play, CheckCircle } from "lucide-react"; const courses
= [ { id: 1, title: "Sustainable Rice Farming", tag: "Beginner", desc: "Learn sustainable and eco-friendly
cultivation techniques for better yields.", modules: 7, duration: "3 Hours", students: 4305, progress: 68,
enrolled: true, }, { id: 2, title: "Water Management Basics", tag: "Beginner", desc: "Efficient irrigation
and water conservation methods for rice cultivation.", modules: 4, duration: "1.5 Hours", students: 2890,
progress: 35, enrolled: true, }, { id: 3, title: "Organic Farming Certification", tag: "Intermediate", desc:
"Complete guide to get organic certification for your farm.", modules: 10, duration: "5 Hours", students:
1530, progress: 0, enrolled: false, }, { id: 4, title: "Advanced Pest Management", tag: "Advanced", desc:
"Natural and integrated pest management strategies using modern techniques.", modules: 8, duration: "4 Days",
students: 1400, progress: 0, enrolled: false, }, ]; const achievements = [ { icon: "■", label: "Quick
Learner", count: 3 }, { icon: "■", label: "Knowledge Seeker", count: 141 }, { icon: "■■■■", label: "Expert
Farmer", count: 0 }, { icon: "■", label: "Certified Farmer", count: 1 }, ]; const TAG_COLORS: Record<string,
string> = { Beginner: "bg-green-100 text-green-700", Intermediate: "bg-yellow-100 text-yellow-700", Advanced:
"bg-red-100 text-red-700", }; export default function LearningPage() { return ( <div className="min-h-screen
bg-gray-50"> <header className="bg-primary-DEFAULT text-white px-6 py-4"> <h1 className="text-xl
font-bold">■ Learning Center</h1> <p className="text-sm text-indigo-200">Expand your farming knowledge</p>
</header> <div className="max-w-7xl mx-auto px-4 py-6"> { /* Stats */ } <div className="grid grid-cols-2
md:grid-cols-4 gap-4 mb-8"> [ { label: "Courses Enrolled", value: "3", icon: BookOpen }, { label:
"Completed", value: "1", icon: CheckCircle }, { label: "Certificates", value: "1", icon: Award }, { label:
"Hours Learned", value: "4.5", icon: Clock },

].map(({ label, value, icon: Icon }) => ( <div key={label} className="bg-white rounded-2xl p-4 shadow-sm
text-center"> <Icon className="mx-auto mb-2 text-primary-DEFAULT" size={22} /> <p className="text-2xl
font-bold text-gray-900">{value}</p> <p className="text-xs text-gray-500 mt-1">{label}</p> </div> )) </div>
{ /* Category Tabs */ } <div className="flex gap-2 overflow-x-auto pb-2 mb-6"> [{"All Courses", "Cultivation",
"Water Management", "Pest Control", "Certification", "Harvest"].map(c => ( <button key={c}
className={` whitespace-nowrap px-4 py-2 rounded-full text-sm font-medium transition ${ c === "All Courses" ?
"bg-primary-DEFAULT text-white" : "bg-white text-gray-600 border border-gray-200" } `}> {c} </button> ))
</div> { /* Courses Grid */ } <div className="grid grid-cols-1 md:grid-cols-2 gap-5 mb-8"> {courses.map(c => (
<div key={c.id} className="bg-white rounded-2xl shadow-sm overflow-hidden hover:shadow-md transition"> <div
className="h-32 bg-gradient-to-br from-green-400 to-green-600 flex items-center justify-center"> <BookOpen
className="text-white opacity-40" size={48} /> </div> <div className="p-5"> <div className="flex items-center
justify-between mb-2"> <span className={` text-xs font-semibold px-2 py-0.5 rounded-full
${TAG_COLORS[c.tag]} `}> {c.tag} </span> <span className="text-xs text-primary-DEFAULT
font-semibold">Enrolled</span> </div> <h3 className="text-base font-bold text-gray-900 mb-1">{c.title}</h3>
<p className="text-sm text-gray-500 mb-3 line-clamp-2">{c.desc}</p> <div className="flex items-center gap-4
text-xs text-gray-400 mb-3"> <span className="flex items-center gap-1"><BookOpen size={12} /> {c.modules}
Modules</span> <span className="flex items-center gap-1"><Clock size={12} /> {c.duration}</span> <span
className="flex items-center gap-1"><Users size={12} /> {c.students.toLocaleString()}</span> </div>
{c.enrolled && c.progress > 0 && ( <div className="mb-3"> <div className="flex justify-between text-xs
text-gray-500 mb-1"> <span>Progress</span><span>{c.progress}%</span> </div> <div className="h-2 bg-gray-100
rounded-full"> <div className="h-2 bg-green-500 rounded-full transition-all" style={{ width: `${c.progress}%`
}} /> </div> </div> )} <button className={` w-full flex items-center justify-center gap-2 py-2.5 rounded-xl
text-sm font-semibold transition ${ c.enrolled ? "bg-primary-DEFAULT text-white hover:bg-primary-dark" :
"border border-primary-DEFAULT text-primary-DEFAULT hover:bg-indigo-50" } `}>

<Play size={14} /> {c.enrolled ? "Continue" : "Start Course"} </button> </div> </div> ))} </div> { /*
Achievements */ } <div className="bg-white rounded-2xl shadow-sm p-6"> <h2 className="font-bold text-gray-900
mb-4">■ Your Learning Achievements</h2>
```

■ `app/api/auth/[...nextauth]/route.ts`

- `store/useStore.ts`

- lib/api.ts

Page 12

```
getReasons:() => api.get("/mobile/order/cancellation-reasons"), }; export default api;
```

A12. Providers Component

■ components/layout/Providers.tsx

```
"use client"; import { SessionProvider } from "next-auth/react"; export function Providers({ children }: {  
children: React.ReactNode }) { return <SessionProvider>{children}</SessionProvider>; }
```

■ PART B — BACKEND (Node.js · Express.js · MongoDB · Mongoose)

B1. Project Configuration

■ package.json

```
{ "name": "nelotsavam-backend", "version": "1.0.0", "description": "Nelotsavam Backend API - Node.js + Express + MongoDB", "type": "module", "main": "src/index.js", "scripts": { "start": "node src/index.js", "dev": "nodemon src/index.js", "test": "node --experimental-vm-modules node_modules/.bin/jest --forceExit" }, "dependencies": { "bcryptjs": "^2.4.3", "compression": "^1.7.4", "cors": "^2.8.5", "dotenv": "^16.0.3", "express": "^4.18.2", "express-rate-limit": "^6.7.0", "express-validator": "^7.0.1", "helmet": "^7.0.0", "jsonwebtoken": "^9.0.0", "moment": "^2.29.4", "mongoose": "^8.0.0", "multer": "^1.4.5-lts.1", "socket.io": "^4.6.1" }, "devDependencies": { "@babel/core": "^7.22.0", "@babel/preset-env": "^7.22.0", "babel-jest": "^29.5.0", "jest": "^29.5.0", "nodemon": "^3.0.0", "supertest": "^6.3.3" } }
```


B3. Auth Middleware

■ `src/services/mobile/middlewares/auth.js`

```
import jwt from "jsonwebtoken"; const auth = (req, res, next) => { const token =
req.headers.authorization?.split(" ")[1]; if (!token) return res.status(401).json({ success: false, message:
"No token provided." }); try { req.user = jwt.verify(token, process.env.JWT_SECRET); next(); } catch { return
res.status(401).json({ success: false, message: "Invalid or expired token." }); } }; export default auth;
```

B4. Auth Module

■ auth/model.js

```
import mongoose from "mongoose"; import bcrypt from "bcryptjs"; const userSchema = new mongoose.Schema({
  name: { type: String, required: true, trim: true }, email: { type: String, required: true, unique: true,
  lowercase: true }, phone: { type: String }, password: { type: String, required: true, select: false },
  district: { type: String }, farmSize: { type: String }, experience: { type: String }, language: { type:
  String, enum: ["en", "ml"], default: "en" }, step_count: { type: Number, default: 0 }, dhanyaPoints: { type:
  Number, default: 0 }, sevasCompleted: { type: Number, default: 0 }, level: { type: Number, default: 1 },
  achievements: [{ type: String }], profile_picture_url: { type: String }, visibility_setting: { type:
  mongoose.Schema.Types.ObjectId, ref: "VisibilitySetting" }, }, { timestamps: true }); userSchema.pre("save",
  async function (next) { if (!this.isModified("password")) return next(); this.password = await
  bcrypt.hash(this.password, 12); next(); }); userSchema.methods.comparePassword = function (plain) { return
  bcrypt.compare(plain, this.password); }; export const User = mongoose.model("User", userSchema);
```

■ auth/controller.js

[illegible]

■ auth/routes.js

```
import { Router } from "express"; import auth from "../middlewares/auth.js"; import { register, login,
getProfile } from "./controller.js"; const router = Router(); router.post("/register", register);
router.post("/login", login); router.get("/profile", auth, getProfile); export default router;
```

B5. Friends Module

■ friends/model.js

```
import mongoose from "mongoose"; const workoutEventSchema = new mongoose.Schema( { workout_date_iso: { type: String, required: true }, // "YYYY-MM-DD" for prefix filtering activity_type: { type: String }, step_count: { type: Number, default: 0 }, calories: { type: Number, default: 0 }, duration_string: { type: String }, distance_string: { type: String }, }, { _id: false } ); const friendSchema = new mongoose.Schema( { userId: { type: mongoose.Schema.Types.ObjectId, ref: "User", required: true }, friendId: { type: mongoose.Schema.Types.ObjectId, ref: "User", required: true }, status: { type: String, enum: ["pending", "accepted", "rejected"], default: "accepted", }, workoutHistory: [workoutEventSchema], }, { timestamps: true } ); const visibilitySettingSchema = new mongoose.Schema( { userId: { type: mongoose.Schema.Types.ObjectId, ref: "User", required: true, unique: true }, visibility_preference: { type: String, enum: ["Everyone", "Only Friends", "Private"], default: "Everyone", }, share_achievements: { type: Boolean, default: true }, share_workout_history: { type: Boolean, default: true }, }, { timestamps: true } ); export const Friend = mongoose.model("Friend", friendSchema); export const VisibilitySetting = mongoose.model("VisibilitySetting", visibilitySettingSchema);
```

■ friends/controller.js

```
import mongoose from "mongoose"; import { Friend, VisibilitySetting } from "../model.js"; // ■■ Add Friend export const addFriend = async (req, res) => { const { friendId } = req.body; const userId = req.user._id; if (!mongoose.Types.ObjectId.isValid(friendId)) { return res.status(400).json({ success: false, message: "Invalid friend ID." }); } if (userId === friendId) { return res.status(400).json({ success: false, message: "You cannot add yourself." }); } try { const existing = await Friend.findOne({ $or: [ { userId, friendId }, { userId: friendId, friendId: userId }, ], }); if (existing) { return res.status(400).json({ success: false, message: "Friendship already exists." }); } const friendship = await Friend.create({ userId, friendId, status: "accepted" }); return res.status(201).json({ success: true, friendship }); } catch (err) { return res.status(500).json({ success: false, message: err.message }); } }; // ■■ Remove Friend export const unfriend = async (req, res) => { const { friendId } = req.body; const userId = req.user._id; try { await Friend.deleteMany({ $or: [ { userId, friendId }, { userId: friendId, friendId: userId }, ], }); return res.json({ success: true, message: "Unfriended successfully." }); } catch (err) { return res.status(500).json({ success: false, message: err.message }); } }; // ■■ Get My Friends (with step data) export const getMyFriends = async (req, res) => { const userId = req.user._id; try { const friendships = await Friend.find({ $or: [{ userId }, { friendId: userId }, status: "accepted", ]).populate("userId friendId", "first_name last_name email profile_picture_url step_count"); return res.json({ success: true, friends: friendships }); } catch (err) { return res.status(500).json({ success: false, message: err.message }); } }; // ■■ Log Daily Workout (idempotent) export const addDailyWorkout = async (req, res) => { const userId = req.user._id; const { friendId, workout_date_iso, activity_type, step_count, calories, duration_string, distance_string } = req.body; if (!mongoose.Types.ObjectId.isValid(friendId)) { return res.status(400).json({ success: false, message: "Invalid friend ID." }); } try { const friendship = await Friend.findOne({ $or: [ { userId, friendId }, { userId: friendId, friendId: userId }, ], status: "accepted", }); if (!friendship) { return res.status(404).json({ success: false, message: "Friendship not found." }); } // Idempotent: replace existing entry for same date friendship.workoutHistory = friendship.workoutHistory.filter( (e) => e.workout_date_iso !== workout_date_iso ); friendship.workoutHistory.push({ workout_date_iso, activity_type, step_count, calories, duration_string, distance_string, }); await friendship.save(); return res.json({ success: true, workoutHistory: friendship.workoutHistory }); } catch (err) { return res.status(500).json({ success: false, message: err.message }); } }; // ■■ Get Monthly Workout History export const getMonthWorkoutHistory = async (req, res) => { const userId = req.user._id; const { friendId, month } = req.query; // month = "YYYY-MM" if (!mongoose.Types.ObjectId.isValid(friendId)) { return res.status(400).json({ success: false, message: "Invalid friend ID." }); } try { const friendship = await Friend.findOne({ $or: [ { userId, friendId },
```

■ friends/routes.js

```
import { Router } from "express"; import auth from "../../middlewares/auth.js"; import { addFriend, unfriend, getMyFriends, addDailyWorkout, getMonthWorkoutHistory, getMySettings, updateMySettings, } from "../controller.js"; const router = Router(); router.post("/add", auth, addFriend); router.delete("/unfriend", auth, unfriend); router.get("/list", auth, getMyFriends); router.post("/workout", auth, addDailyWorkout);
```

```
router.get("/workout", auth, getMonthWorkoutHistory); router.get("/settings", auth, getMySettings);  
router.put("/settings", auth, updateMySettings); export default router;
```


B7. Activity Session Module

■ activity-session/model.js

```
import mongoose from "mongoose"; const routePointSchema = new mongoose.Schema( { latitude: { type: Number,
required: true }, longitude: { type: Number, required: true }, timestamp: { type: Number }, // Unix ms }, {
_id: false } ); const activitySessionSchema = new mongoose.Schema( { user: { type:
mongoose.Schema.Types.ObjectId, ref: "User" }, type: { type: String }, // validated against ActivityType
dropdown targetKm: { type: Number, default: 0 }, targetType: { type: String }, // validated against
TargetType dropdown startTime: { type: Date }, endTime: { type: Date }, durationMin: { type: Number },
distanceCovered: { type: Number }, steps: { type: Number, default: 0 }, caloriesBurned: { type: Number },
avgSpeed: { type: Number }, route: [routePointSchema], }, { timestamps: true } ); // Dropdown collections
const activityTypeSchema = new mongoose.Schema({ name: { type: String, required: true, unique: true,
lowercase: true }, displayName: { type: String, required: true }, }); const targetTypeSchema = new
mongoose.Schema({ name: { type: String, required: true, unique: true, lowercase: true }, displayName: { type:
String, required: true }, unit: { type: String, required: true }, }); export const ActivitySession =
mongoose.model("ActivitySession", activitySessionSchema); export const ActivityType =
mongoose.model("ActivityType", activityTypeSchema); export const TargetType = mongoose.model("TargetType",
targetTypeSchema);
```

- **activity-session/controller.js**

[illegible]

- `activity-session/routes.js`

```
import { Router } from "express"; import auth from "../../middlewares/auth.js"; import { startSession, endSession, getSessionReport, addActivityType, addTargetType, } from "./controller.js"; const router = Router(); // Session lifecycle router.post("/start", auth, startSession); router.patch("/:sessionId/end", auth, endSession); router.get("/:sessionId/report", auth, getSessionReport); // Admin dropdown management router.post("/admin/activity-types", auth, addActivityType); router.post("/admin/target-types", auth, addTargetType); export default router;
```

B8. Order Module

- `order/model.js`

```
import mongoose from "mongoose"; const mealItemSchema = new mongoose.Schema( { name: String, price: Number, quantity: Number, }, { _id: false } ); const orderSchema = new mongoose.Schema( { userId: { type: mongoose.Schema.Types.ObjectId, ref: "User", required: true }, orderNumber: { type: Number, unique: true }, meals: [mealItemSchema], amount: { type: Number, default: 0 }, discount: { type: Number, default: 0 }, deliveryFee: { type: Number, default: 0 }, platformFee: { type: Number, default: 0 }, status: { type: String, enum: ["Placed", "Accepted", "Preparing", "Ready", "Delivered", "Cancelled"], default: "Placed" }, cancelReason: { type: String }, cancellationFee: { type: Number, default: 0 }, isCancelled: { type: Boolean, default: false }, paymentMethod: { type: String }, deliveryAddress: { type: String }, }, { timestamps: true } ); // Auto-incrementing order counter const counterSchema = new mongoose.Schema( { name: { type: String, required: true, unique: true }, value: { type: Number, default: 1000 }, }, { timestamps: true } ); const cancellationReasonSchema = new mongoose.Schema( { reason: { type: String, required: true, unique: true }, }, { timestamps: true } ); const Order = mongoose.model("Order", orderSchema); export const Counter = mongoose.model("Counter", counterSchema); export const CancellationReason = mongoose.model("CancellationReason", cancellationReasonSchema);
```

- **order/controller.js**

[illegible]

- `order/routes.js`

```
import { Router } from "express"; import auth from "../../middlewares/auth.js"; import { placeOrder,
getUserOrders, getOrderById, updateOrderStatus, cancelOrder, repeatOrder, getCancellationReasons,
addCancellationReason, } from "./controller.js"; const router = Router(); router.post("/", auth, placeOrder);
router.get("/", auth, getUserOrders); router.get("/cancellation-reasons", getCancellationReasons);
router.post("/cancellation-reasons", auth, addCancellationReason); router.get("/:orderId", auth,
getOrderById); router.patch("/:orderId/status", auth, updateOrderStatus); router.post("/:orderId/cancel",
auth, cancelOrder); router.post("/:orderId/repeat", auth, repeatOrder); export default router;
```

B9. Unit Tests (Jest)

■ tests/order.test.js

```
/** * Order Module - Unit Tests * Run: npm test */ describe("Order Controller - Business Logic", () => { //
Helper: build a mock order object const mockOrder = (overrides = {}) => ({ amount: 330, status: "Placed",
meals: [{ name: "Grilled Chicken Bowl", price: 280, quantity: 1 }], deliveryFee: 30, platformFee: 20,
...overrides, }); test("cancellation fee is 45 rupees", () => { const CANCELLATION_FEE = 45; const order =
mockOrder({ amount: 330 }); const refund = order.amount - CANCELLATION_FEE; expect(refund).toBe(285); });
test("displayOrderId is zero-padded with hash prefix", () => { const orderNumber = 1001; const displayOrderId
= `#${String(orderNumber).padStart(4, "0")}`; expect(displayOrderId).toBe("#1001"); }); test("delivered
order cannot be cancelled", () => { const order = mockOrder({ status: "Delivered" }); const canCancel =
order.status !== "Delivered"; expect(canCancel).toBe(false); }); test("valid status list does not include
Placed or Cancelled", () => { const VALID_STATUSES = ["Accepted", "Preparing", "Ready", "Delivered"];
expect(VALID_STATUSES).not.toContain("Placed"); expect(VALID_STATUSES).not.toContain("Cancelled"); });
test("repeat order copies correct fields", () => { const original = mockOrder(); const repeated = { meals:
original.meals, amount: original.amount, deliveryFee: original.deliveryFee, platformFee:
original.platformFee, status: "Placed", }; expect(repeated.status).toBe("Placed");
expect(repeated.meals).toEqual(original.meals); }); });

describe("Water Intake - Period Logic", () => { const VALID_PERIODS = ["today", "week", "month", "year"];
test("rejects invalid period string", () => { const period = "yesterday";
expect(VALID_PERIODS.includes(period)).toBe(false); }); test("accepts all valid period strings", () => {
VALID_PERIODS.forEach((p) => expect(VALID_PERIODS.includes(p)).toBe(true)); }); }); describe("Friends Module
- Workout History (idempotent)", () => { test("replaces existing entry for same date", () => { const history
= [ { workout_date_iso: "2026-05-01", step_count: 5000 }, { workout_date_iso: "2026-05-02", step_count: 3000
}, ]; const newEntry = { workout_date_iso: "2026-05-01", step_count: 7500 }; const updated = history
.filter((e) => e.workout_date_iso !== newEntry.workout_date_iso) .concat(newEntry);
expect(updated).toHaveLength(2); const entry = updated.find((e) => e.workout_date_iso === "2026-05-01");
expect(entry.step_count).toBe(7500); }); });
```

C1. Full API Endpoint Reference

Method	Endpoint	Description
POST	/api/auth/register	Register new farmer
POST	/api/auth/login	Login and get JWT
GET	/api/auth/profile	Get authenticated user profile
POST	/api/mobile/friends/add	Add a friend
DELETE	/api/mobile/friends/unfriend	Remove a friend
GET	/api/mobile/friends/list	List all friends with step data
POST	/api/mobile/friends/workout	Log shared workout (idempotent)
GET	/api/mobile/friends/workout	Get monthly workout history
GET	/api/mobile/friends/settings	Get visibility settings
PUT	/api/mobile/friends/settings	Update visibility settings
POST	/api/mobile/water-intake	Log new water intake record
GET	/api/mobile/water-intake?of=today	Query by period (today/week/month/year)
GET	/api/mobile/water-intake/range	Query by custom date range
PUT	/api/mobile/water-intake/:id	Update intake record
DELETE	/api/mobile/water-intake/:id	Delete intake record
POST	/api/mobile/activity/start	Start activity session
PATCH	/api/mobile/activity/:id/end	End session & submit metrics
GET	/api/mobile/activity/:id/report	Get post-session report
POST	/api/mobile/activity/admin/activity-types	Add activity type (admin)
POST	/api/mobile/activity/admin/target-types	Add target type (admin)
POST	/api/mobile/order	Place new order
GET	/api/mobile/order	Get all user orders
GET	/api/mobile/order/:orderId	Get single order by ID
PATCH	/api/mobile/order/:orderId/status	Update order status
POST	/api/mobile/order/:orderId/cancel	Cancel order with reason
POST	/api/mobile/order/:orderId/repeat	Repeat a previous order
GET	/api/mobile/order/cancellation-reasons	Get valid cancellation reasons

C2. Project Setup Instructions

Backend Setup

```
# 1. Navigate to backend folder cd nelotsavam-backend # 2. Install dependencies npm install # 3. Create environment file cp .env.example .env # Edit .env: # MONGO_URI=mongodb://localhost:27017/nelotsavam # JWT_SECRET=your_super_secret_key_here # PORT=3001 # CORS_ORIGIN=http://localhost:3000 # 4. Run in development mode (auto-reload) npm run dev # 5. Run tests npm test # 6. Build Docker image (optional) docker build -t nelotsavam-backend . docker run -p 3001:3001 --env-file .env nelotsavam-backend
```

Frontend Setup

```
# 1. Navigate to frontend folder cd nelotsavam-frontend # 2. Install dependencies npm install # 3. Create environment file cp .env.example .env.local # Edit .env.local: # NEXT_PUBLIC_API_URL=http://localhost:3001/api # NEXTAUTH_SECRET=your_nextauth_secret_here # NEXTAUTH_URL=http://localhost:3000 # BACKEND_URL=http://localhost:3001 # 4. Run in development mode npm run dev # App will be available at http://localhost:3000 # 5. Build for production npm run build npm start
```

GitHub Upload

```
# Push BACKEND to GitHub cd nelotsavam-backend git init git add . git commit -m "feat: Nelotsavam backend - MCA major project (Vivek B K 20242MCA0340)" git branch -M main git remote add origin https://github.com/<username>/nelotsavam-backend.git git push -u origin main # Push FRONTEND to GitHub cd nelotsavam-frontend git init git add . git commit -m "feat: Nelotsavam frontend - Next.js 15 (MCA major project)" git branch -M main git remote add origin https://github.com/<username>/nelotsavam-frontend.git git push -u origin main
```